

Marketing Mix Model Metrics Simplified

Coefficient & Decomposition Metrics

- **rn coef**: Simple relationship strength between marketing activity and sales
- **xDecompAgg**: Total sales impact from each marketing channel
- **xDecompPerc**: What percentage of sales each channel contributes
- **xDecompMeanNon0**: Average sales impact when a channel is active
- **xDecompMeanNon0Perc**: Percentage impact when channel is active
- **RF variants**: Above metrics adjusted for response curves

Model Performance Metrics

- **pos**: Ranking of this model solution compared to others
- **train_size**: Percentage of data used for training
- **rsq_train/val/test**: How well the model fits the data (higher is better)
- **nrmse**: Error measurement (lower is better)
- **decomp.rssd**: How well decomposition performs
- **mape**: Average percentage error

Regularization Parameters

- **lambda** variations: Technical settings that prevent overfitting

Solution Identifiers

- **solID/trial/iter**: Technical identifiers for the model run
- **robynPareto**: Indicates if this is an optimal solution

Financial Metrics

- **total_spend**: Total marketing investment
- **mean_spend**: Average spend per channel
- **mean_spend_adstocked**: Average spend adjusted for lingering effects
- **mean_carryover**: Lasting impact of marketing
- **spend_share**: How budget is distributed across channels
- **effect_share**: How sales impact is distributed across channels
- **roi_mean/total**: Return on investment metrics
- **cpa_total**: Cost to acquire one customer/sale

Statistical & Validation Metrics

- **boot_mean/boot_se**: Statistical validation measurements
- **ci_low/ci_up**: Confidence range for predictions
- **carryover_pct**: Percentage of effect that continues over time

How To Use These Metrics

1. **Budget Allocation**: Use effect_share and ROI to redistribute marketing spend
2. **Channel Effectiveness**: Compare xDecompPerc across channels
3. **Model Reliability**: Check rsq and nrmse values for model accuracy
4. **Long-term Impact**: Evaluate carryover_pct for sustained effects
5. **Optimization**: Focus on channels with high effect_share but low spend_share